Broadcast-Gain: Two-Byte Junction Control under Partial Information

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Abstract

Broadcast-Gain (BG) uses two-byte reports of nearby demand to schedule a junction under partial observation. Its uncertainty-dependent switching rule has an unexpected consequence: with an empty served axis and one waiting agent, improving report delivery can postpone the next switch from 24 to 48 steps. We derive the expected switch time and replace the conditional fallback with an unconditional deadline. In the high-load, high-loss workload, this correction reduces 95th-percentile waiting from 173.4 to 164.9 steps and increases throughput from 678.9 to 682.8 crossings per 1,000 steps. Phase traces show that 46.0% of original phases exceed the 24-step fallback. The correction eliminates these extensions, including the many phases held on inconclusive reports with support above the strictly low-support range. A narrower intervention confined to that range accounts for a small part of the improvement. A separately selected 12-step timer reaches 156.8 waiting steps; its advantage disappears when clearance becomes expensive or demand is persistently unbalanced.

Keywords: junction control; sparse communication; partial observation; service deadlines.

1 Introduction

At a junction, frequent phase changes interrupt service, while long phases leave agents waiting on the other approaches. Reports from nearby agents can help the controller decide when to switch, but silence is ambiguous: an approach may be empty, or its reports may have been lost. A fallback timer provides a schedule when those reports are unavailable.

Broadcast-Gain (BG) uses two-byte reports to score the competing approaches. It discounts old reports, requires stronger evidence when few reports are available, and lengthens its minimum phase under low support. Each choice is intended to avoid unnecessary switching. Together, however, they interact awkwardly with the fallback: an empty cache invokes a four-cycle timer, while a nonempty cache can keep the phase active for eight cycles. A report can therefore disable the shorter deadline while supplying too little evidence to authorize a switch.

We start with one agent waiting on the unserved axis and no traffic on the served axis. Erasing all its reports produces a switch after four cycles; delivering a fresh report every cycle extends the wait to eight. The report is timely and correctly identifies demand. The delay arises when its arrival suppresses the fallback, an interaction we analyze by deriving the expected switch time under independent losses.

We then make the four-cycle fallback unconditional and compare the corrected controller with the original rule. To assess how much of the workload improvement comes from the strictly low-support state in the example, we also apply a narrower patch to that branch alone. Phase-duration histograms and switch-reason counts let us follow both changes through the controller's operation, alongside comparisons with independently selected timers.

The workloads use one junction with fixed routes. We use greedy movement as the primary control because legal movement carries no energy or risk penalty, and examine learned movement and message contents in secondary comparisons. Sensitivity tests vary clearance cost and demand imbalance to assess how the choice of deadline affects waiting and throughput.

Maximum green times and phase recall are established signal-control mechanisms: an actuated maximum-green timer can depend on a conflicting call, while recall requests a phase even without detection [1]. BG's corrected

deadline runs regardless of a detected opposing call or surviving reports. Pressure control allocates service from queue measurements [2]; for our packet-pressure comparison, we use BG's incomplete local reports to compute the advantage.

2 Junction model and Broadcast-Gain

2.1 Movement, reports, and service

Four 48-cell directed lanes feed a single gate, with opposing lanes forming an axis. The controller alternates between axes $S \in \{-1, +1\}$. Each switch closes the gate for c environment steps before the new axis opens; we include this clearance interval in the *phase duration*. The main workload uses $c = 2$, and later tests vary the clearance cost. Agents attempt reports every C steps on a calendar fixed in physical time.

Agents choose MOVE or WAIT. Movement into an occupied cell, or through a red or clearing gate, is masked. Occupancy is checked before simultaneous movement, so an agent cannot enter a cell vacated in the same step. When both lanes of the served axis attempt a crossing, an interlock admits the agent with greater accumulated waiting, breaking ties by identifier. At most one agent crosses per step. This interlock is identical for all methods, and its tie-breaking information is not supplied to the scheduler.

Greedy movement chooses MOVE whenever legal. The learned alternative uses a frozen proximal policy optimization (PPO) actor, either sampling an action or deterministically selecting its larger legal logit. A reliable local interface publishes phase and clearance status from one authoritative controller; loss applies to the agent reports.

An agent within eight cells of the gate reports its axis s_i, distance d_i, and a waiting-progress residual:

$$x_i = \mathrm{clip}\Big(\frac{w_i}{64} - \bar{v}_i, -1, 1\Big), \qquad z_i = \mathrm{round}(127 x_i). \tag{1}$$

Here w_i includes waiting in the approach's input buffer, and $\bar{v}_i$ is the fraction of the preceding C steps with accepted forward progress. One signed byte stores z_i and a second byte stores axis and distance. Sender identity and sequence number are transport fields outside this payload. Reports are not acknowledged or retransmitted.

The controller keeps the newest report per sender, removing it at age $2C$ or when the sender is observed crossing. For report age a_j, define freshness $\eta_j = e^{-a_j/C}$ and decoded residual $\hat{x}_j = z_j/127$. The scores and support are

$$Q_s = \sum_{j:s_j=s} \eta_j e^{-d_j/8} \frac{1+\hat{x}_j}{2}, \qquad n_{\mathrm{eff}} = \sum_j \eta_j, \qquad u = \min(1, n_{\mathrm{eff}}/8). \tag{2}$$

Each cached distance stays at its reported value until a replacement arrives or the report is removed. Support summarizes the amount and age of received information; the controller has no count of missing reports with which to estimate queue coverage.

2.2 The original switching rule

Let h be the number of complete reporting cycles since the last switch. The unserved axis's normalized advantage and BG's minimum hold are

$$a_S = \frac{Q_{-S} - Q_S}{Q_{+1} + Q_{-1} + 1}, \tag{3}$$

$$m(u) = \max\{2 + \lceil 2(1-u) \rceil, \lceil c/C \rceil + 1\}. \tag{4}$$

With a nonempty cache, BG waits until $h \geq m(u)$, then switches if

$$u a_S > 0.12 + 0.20(1-u) \quad \text{or} \quad h \geq 8. \tag{5}$$

With an empty cache, it switches at $h \geq 4$ instead. Ties hold the phase until a duration limit applies. Cache updates precede the decision; a switch resets phase age and begins clearance immediately. BG considers switches only at report boundaries.

BG scheduling denotes this rule with greedy movement and no logit adjustment. *Full BG* also applies a report-conditioned move-logit increase of at most 0.4 to the frozen PPO actor near an open gate. Appendix A.3 specifies that adjustment and the learned-policy controls; the analysis below uses the scheduling rule alone.

3 Partial information can postpone the fallback

3.1 The single-reporter failure

The scores are nonnegative, so $a_S < 1$. An evidence-triggered switch therefore requires

$$u > 0.12 + 0.20(1 - u) \quad \Longleftrightarrow \quad u > \frac{4}{15} \quad \Longleftrightarrow \quad n_{\text{eff}} > \frac{32}{15}. \tag{6}$$

This condition is necessary, not sufficient. One or two effective reports cannot trigger the evidence branch, even when every report favors the unserved axis.

Consider a phase starting at time zero with an empty served axis and one stationary agent at the opposite gate. No other agents arrive. If all reports are erased, the cache stays empty and BG switches after four cycles. If the agent supplies a fresh report every cycle, $u = 1/8$. Its reports keep the cache nonempty but never pass the evidence test, so BG waits eight cycles. At $C = 6$, the switch moves from step 24 to step 48. After the two-step clearance, the waiting axis first opens at step 26 or step 50, respectively.

Let each report be delivered independently with probability $q = 1 - p$. With one sender, the cache is empty at a boundary exactly when the current and preceding attempts were both lost. Before cycle eight, the next switch is the first such boundary at or after cycle four. If $f_n(q)$ is the probability that n attempts contain no adjacent losses, then

$$f_0 = f_1 = 1, \qquad f_n = q f_{n-1} + q(1 - q) f_{n-2}, \qquad \mathbb{E}[\tau] = C\left(4 + \sum_{n=2}^{5} f_n(q)\right), \tag{7}$$

where τ is the next switch time. At $q = 0.30$ and $C = 6$, the expectation is 31.38 steps. It rises from 24 under complete loss to 48 under perfect delivery (Figure 1). Turning a loss into a delivery can only remove an empty-cache opportunity in this example. Coupling the delivery indicators with common uniforms therefore proves that the expected switch time is nondecreasing in q. Appendix B gives the recurrence and an exhaustive enumeration check.

3.2 A deadline that reports cannot postpone

When a phase starts at t_0, assign it a deadline $t_0 + H$, with integer $H > c$ measured in environment steps. Reports may trigger an earlier switch, but cannot extend the deadline. Let $E(t)$ denote BG's eligible evidence event: t is a report boundary, $u > 0$, $t - t_0 \geq Cm(u)$, and the inequality in (5) holds. The corrected controller, denoted BG-D(H), switches at

$$\tau = \min\{t_0 + H,\ \inf\{t > t_0 : E(t)\}\}. \tag{8}$$

The deadline replaces both the empty-cache fallback and the eight-cycle maximum. It is checked every environment step, outside the support and minimum-hold tests (Algorithm 1). Evidence thresholds, reports, and cache handling are otherwise unchanged.

Proposition 1 (Bounded interruption of service)**.** *For every report sequence, BG-D(H) switches no later than $t_0 + H$. Each axis's uninterrupted interval without green is at most $H + c$, and each phase provides at least one non-clearance step.*

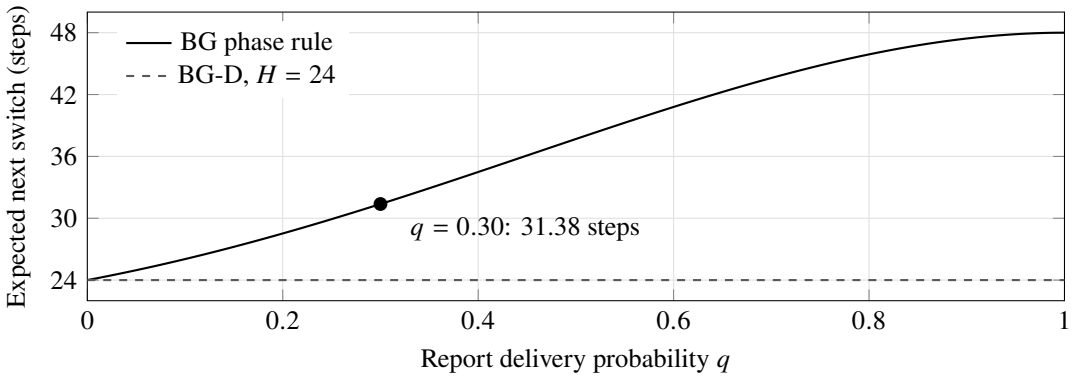


Figure 1. Exact next-switch time for the single-reporter example. The served axis is empty, one agent waits at the opposite gate, and $C = 6$. More reliable delivery keeps a weak cache alive and delays the original rule's next switch. An unconditional deadline at 24 steps gives the same switch time for every delivery pattern. Clearance follows the switch in both cases.

Algorithm 1 BG with an unconditional physical-time deadline

Require: Phase S, start t_0, duration $H > c$, report interval C
At step t, expire old reports and remove senders observed crossing.
Set $E \leftarrow$ false.
if t is a report boundary **then**
    Insert received reports; compute $Q_s, u, a_S, m(u)$.
    $E \leftarrow [u > 0] \wedge [t - t_0 \geq Cm(u)] \wedge [ua_S > 0.12 + 0.20(1 - u)]$.
end if
if $t \geq t_0 + H$ or E **then**
    $S \leftarrow -S$; $t_0 \leftarrow t$; close the gate for the next c steps.
end if
Publish phase and clearance status; apply the common movement interlock.

Proof. The deadline condition forces a switch at $t_0 + H$ unless evidence has already caused one. An axis that loses green at t_0 is selected again by $t_0 + H$ and opens after another c steps. A deadline-triggered phase lasts $H > c$ steps, while the minimum in $E(t)$ also exceeds clearance. Thus the return to green supplies at least one open step. □

For the single reporter, $H = 24$ preserves the original no-report switch time for every delivery pattern, giving the horizontal line in Figure 1.

3.3 Unequal loss also biases the scores

Unequal delivery rates affect the scores used for early switching. Consider a stationary sender whose score contribution is unchanged over two report cycles. Its retained report is current with probability q, one cycle old with probability $(1 - q)q$, and otherwise absent, giving expected freshness weight

$$\mathbb{E}[\eta \mathbf{1}\{\text{cached}\}] = q\left[1 + (1 - q)e^{-1}\right]. \tag{9}$$

The expected weight is 0.488 at $q = 0.40$ and 0.259 at $q = 0.20$, so identical stationary queues can contribute different expected scores. These weights remain unchanged by the deadline correction and can influence switches before it expires. Section 5 examines the resulting service differences under axis-dependent loss.

4 Evaluation

4.1 Workloads, comparisons, and measurements

The main grid crosses populations $N \in \{100, 120, 140\}$, independent erasure probabilities $p \in \{0.60, 0.65, 0.70\}$, and intervals $C \in \{3, 5, 6\}$. These 27 conditions share one layout. After crossing, an agent takes an uncongested return journey lasting an integer number of steps drawn uniformly from 64 to 96, then joins an approach's first-in, first-out buffer. Every 256 steps, the probability of assignment to the horizontal axis alternates between 0.75 and 0.25. Directions within an axis are equally likely. Population is fixed, so later arrival times depend on earlier service.

Our simplest comparison is a fixed timer that alternates every H physical steps without queue reports; separate development workloads select $H = 12$. A boundary-constrained version uses $C\lceil 12/C \rceil$ so that we can assess the effect of permitting switches between reports. Packet-pressure uses BG's cache and advantage with fixed minimum hold, threshold, and maximum hold: $(m, \theta, M) = (3, 0.24, 6)$ cycles. It retains the four-cycle empty-cache fallback. Its development search covers a broader range of holds and thresholds than is obtained by deleting BG's support terms alone.

We evaluate two deadline settings. BG-D($4C$) keeps each condition's original empty-cache fallback duration while making it unconditional, giving a matched intervention on the fallback logic. BG-D(12) uses a physical duration selected independently of C. For the narrower *weak-cache patch*, we extend the four-cycle fallback to every state with $u \le 4/15$; the original rule is unchanged above that threshold.

For a visit ready at b_i and completed at e_i, waiting is

$$W_i = e_i - b_i - 48. \tag{10}$$

This subtracts free-flow traversal while retaining buffer and on-grid delay. Each realization has 10,000 warm-up steps, a 100,000-step scoring window, and 20,000 follow-up steps during which traffic continues unchanged. We include every visit that becomes ready in the scoring window. A visit unfinished after follow-up is assigned infinite waiting; all visits finish in the reported outcomes. Throughput counts scoring-window crossings per 1,000 steps.

Each condition has 20 blocks of four realizations, with a different training seed per block for learned policies. We compute p95 within each realization, average within blocks, and then average across blocks; throughout the paper, "p95 waiting" refers to this mean of realization-level quantiles. Methods share initial states and indexed random inputs for routes, returns, erasures, and action sampling. The indexing preserves pairing as the traffic trajectories diverge.

We report direct paired 95% percentile bootstrap intervals from 10,000 resamples of the 20 blocks, preserving each block's workloads and conditions. Blocks are the resampling units [3], and grid averages weight conditions equally. Controller selection uses separate seeds and workloads at $(N, p, C) = (110, 0.50, 4)$, after which parameters stay fixed for every test condition. Appendix A gives the step order, selection grids, and diagnostic definitions.

4.2 The corrected controller in the workload comparisons

At $N = 140$, $p = 0.70$, and $C = 6$, making the 24-step fallback unconditional lowers p95 from 173.4 to 164.9 steps while leaving the report format and evidence parameters unchanged (Table 1). The paired change is -8.5 steps, with interval $[-9.8, -7.2]$. Throughput also increases, by 3.9 crossings per 1,000, with interval $[3.3, 4.5]$.

The 12-step timer performs better in this workload: BG-D(24) adds 8.1 waiting steps relative to it, with interval $[6.7, 9.4]$, and loses 0.8 crossings per 1,000, with interval $[-1.5, -0.1]$. At $C = 6$, BG-D(12) follows exactly the same schedule as the timer because its minimum evidence-triggered hold is at least 12 steps.

Across the grid, BG-D($4C$) lowers waiting relative to original BG by 2.8 steps on average, with interval $[2.0, 3.6]$ for the reduction (Table 2); its position relative to the 12-step timer varies with the reporting interval.

Table 1. Scheduling at $N = 140$, $p = 0.70$, $C = 6$, and $c = 2$, with greedy movement. Differences are relative to the 12-step fixed timer. Throughput is crossings per 1,000 steps. BG-D(12) is exactly the 12-step timer in this condition, so it shares the reference row.

Scheduler	p95 wait	Throughput	Difference [95% interval]
Fixed 12 / BG-D(12)	156.8	683.6	Reference
Fixed 24	184.6	653.8	+27.8 [26.5, 29.1]
Packet-pressure	171.2	682.1	+14.4 [12.4, 16.3]
Original BG	173.4	678.9	+16.6 [15.5, 17.7]
BG, weak-cache patch	172.7	679.3	+15.9 [14.9, 16.9]
BG-D(24)	164.9	682.8	+8.1 [6.7, 9.4]

BG-D(12) has a smaller grid-average improvement over that timer, 0.7 steps with interval $[0.5, 0.9]$. Earlier evidence-triggered switches account for this difference at $C = 3$ and $C = 5$, while the schedules coincide at $C = 6$.

Table 2. Main-grid comparison with greedy movement. Each row averages nine population-loss conditions. "Boundary" denotes the timer with duration $C\lceil 12/C \rceil$. BG-D($4C$) preserves the original no-report duration; BG-D(12) keeps its deadline fixed in physical time.

C	Fixed 12	Boundary	Pressure	BG	BG-D($4C$)	BG-D(12)
p95 waiting, steps						
3	128.3	128.3	127.4	127.1	126.6	126.6
5	128.3	133.1	133.8	134.1	130.7	128.0
6	128.3	128.3	138.5	139.9	135.4	128.3
Throughput, crossings per 1,000 steps						
3	608.5	608.5	609.6	610.0	610.2	610.2
5	608.5	606.6	607.7	607.2	608.3	608.7
6	608.5	608.5	605.2	603.3	605.7	608.5

The pressure comparison changes appreciably with the development search. Restricting it to short holds and low thresholds gives stress-case p95 176.8, above original BG's 173.4; expanding the search lowers it to 171.2. Across the grid, original BG minus the expanded-grid pressure rule is +0.4 waiting steps, with interval $[-0.4, 1.2]$. We use the expanded search in the main tables.

4.3 Phase extensions and the weak-cache intervention

To follow the intervention through the controller, we examine phase durations and the conditions recorded at each decision. In the stress workload, 46.0% of original BG phases extend beyond the four-cycle fallback and 10.0% reach the eight-cycle maximum (Figure 2). With BG-D(24), every phase ends by 24 steps and mean duration falls from 28.4 to 21.5 steps. The deadline triggers 69.0% of switches; evidence triggers the remaining 31.0%.

The low-support regime used in the counterexample is rare in this workload. A nonempty cache with $n_{\text{eff}} \leq 32/15$ occurs at 0.9% of report boundaries (Table 3) and accounts for 1.6% of suppressed four-cycle fallbacks. Most extensions occur with support above that range and an advantage below the switching threshold.

We use the weak-cache patch to isolate the branch implicated by the single-reporter calculation. Extending the fallback to states where the evidence test cannot pass reduces p95 by 0.7 steps, with interval $[0.2, 1.2]$. The unconditional deadline produces the larger 8.5-step reduction by also reaching phases held on inconclusive, better-supported reports. Both interventions change subsequent traffic, so their waiting-time reductions describe separate closed-loop comparisons.

Unused green falls from 8.7% to 5.0% of non-clearance steps. We count it from the traffic trace whenever an agent waits at the unserved gate and both served lanes are empty at their gates. This is an offline diagnostic.

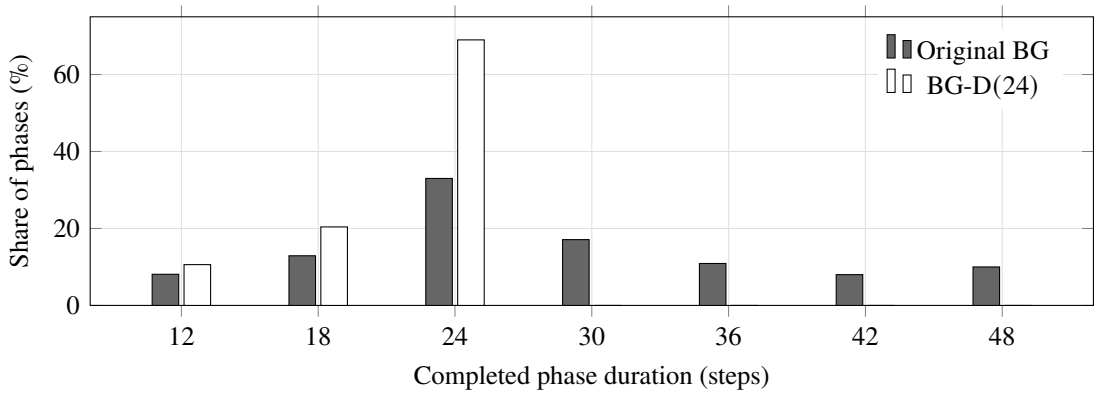


Figure 2. Phase durations in the stress workload, including clearance. The original controller can continue beyond its 24-step no-report fallback whenever the cache is nonempty. The deadline intervention removes those extensions while retaining earlier evidence-triggered switches. Each histogram is normalized within realizations before block averaging.

Table 3. Controller diagnostics in the stress workload. Switch reasons partition all switches, and support classes partition report boundaries. The final two rows summarize extended phases and unused green. Appendix A.4 specifies the denominator for each measure.

Diagnostic (%)	Original BG	BG-D(24)
Reason for switching		
Evidence	89.9	31.0
Empty-cache fallback	0.1	0.0
Eight-cycle maximum	10.0	0.0
Unconditional deadline	0.0	69.0
Support at report boundaries		
Empty: $n_{\text{eff}} = 0$	0.1	0.1
Weak: $0 < n_{\text{eff}} \leq 32/15$	0.9	1.2
Intermediate: $32/15 < n_{\text{eff}} < 8$	35.8	38.0
Support at its cap: $n_{\text{eff}} \geq 8$	63.2	60.7
Phases longer than 24 steps	46.0	0.0
Unused green with opposing gate demand	8.7	5.0

5 Sensitivity to channel quality, demand, and clearance

5.1 Different loss rates on the two axes

The asymmetric-channel test assigns erasure probabilities 0.60 and 0.80 to the axes and repeats each realization with those labels exchanged. Routing is unchanged; the lower-loss axis accounts for between 49.9% and 50.2% of new visits across the compared controllers. Original BG gives that axis 58.0% of non-clearance service time and a 37.3-step advantage in p95 waiting (Table 4).

Table 4. Axis-dependent erasures at $N = 140$, $C = 6$, and $c = 2$, with greedy movement. "Green share" is the fraction of non-clearance time assigned to the lower-loss axis. The waiting gap is higher-loss minus lower-loss p95. Loss labels are exchanged within every paired realization.

Scheduler	Lower-loss p95	Higher-loss p95	Green share	Gap [95% interval]
Fixed 12 / BG-D(12)	156.8	156.8	50.0%	0.0 (exact)
Original BG	166.1	203.4	58.0%	37.3 [34.7, 39.9]
BG-D(24)	160.8	178.6	53.6%	17.8 [15.3, 20.5]

BG-D(24) reduces the waiting gap to 17.8 steps and the lower-loss green share to 53.6%. Waiting on the higher-loss axis falls by 24.8 steps, with interval $[23.5, 26.1]$ for the reduction. The remaining gap is consistent with biased evidence influencing switches before the deadline. Because the 12-step timer's trajectory is independent of erasures, exchanging loss labels gives exactly equal axis means; BG-D(12) shares that schedule at this reporting interval.

5.2 Deadline duration under costly clearance and uneven demand

Starting from the stress workload, we vary one factor at a time and keep the controller parameters fixed (Table 5). The variants cover stationary balanced demand, a persistent 90:10 axis split, faster alternation, clearance durations of one and six steps, and populations of 80 and 180. Comparing the 12-step and 24-step timers shows how the duration choice fares as these conditions change.

Table 5. One-factor workload sensitivity at $p = 0.70$ and $C = 6$. All methods use greedy movement. Each pair of columns gives p95 waiting in steps and throughput per 1,000 steps (rate). The reference has $N = 140$, $c = 2$, and a 75:25 axis preference alternating every 256 steps. All other factors remain at those values.

	Fixed 12		Fixed 24		BG		BG-D(24)	
Change	p95	Rate	p95	Rate	p95	Rate	p95	Rate
Reference	156.8	683.6	184.6	653.8	173.4	678.9	164.9	682.8
Steady 50:50	143.5	691.2	164.7	666.3	149.8	697.0	141.9	699.5
Steady 90:10	307.2	443.8	283.0	485.4	214.5	632.0	263.7	546.2
64-step alternation	171.6	679.8	203.7	641.1	190.2	661.5	178.4	675.3
Clearance $c = 1$	139.7	714.2	167.8	685.9	154.6	711.8	146.8	715.7
Clearance $c = 6$	280.4	477.2	181.6	655.1	202.3	663.8	192.7	641.6
Population $N = 80$	77.6	476.8	91.4	454.8	96.3	454.6	82.9	470.8
Population $N = 180$	211.6	757.5	244.8	731.4	237.9	749.8	222.4	756.2

Clearance cost reverses the main timing comparison. When clearance takes six steps, a 12-step phase spends half its time closed, imposing a throughput ceiling of 500 crossings per 1,000 steps. P95 waiting then reaches 280.4 for the 12-step timer, compared with 181.6 for the 24-step timer. BG-D(24) reaches 192.7, adding 11.1 steps relative to the longer timer, with interval $[9.0, 13.3]$; its evidence-triggered switches can incur clearance more frequently than a fixed 24-step phase.

Persistent 90:10 demand favors long phases on the busier axis. Original BG gives lower waiting and higher throughput than either equal-split timer. Capping it at 24 steps increases p95 by 49.2 steps, with interval $[45.6, 53.6]$, and reduces throughput. The additional returns to the lightly loaded axis enforce the service bound at a substantial capacity cost for the dominant queue.

In the faster-alternation, light-population, and heavy-population comparisons, the 12-step timer continues to give lower waiting. Under stationary balanced demand, BG-D(24) has a small mean advantage, reaching 141.9 steps against the timer's 143.5.

6 Discussion and conclusion

The single-reporter calculation gave us a simple way to isolate the interaction between support and fallback timing. The workload traces broaden that account: extended phases usually contain enough reports to clear the support threshold, but their combined advantage remains too small to trigger a switch. That helps explain why the unconditional deadline changes waiting more than the weak-cache patch.

For this class of controller, we would set a service deadline in physical time, enforce it independently of report reception, and compare candidate durations against tuned fixed timers. The 24-step intervention preserves the

original no-report schedule and improves the main high-load comparison with the report and evidence rules held fixed. Choosing its duration still requires workload-specific evaluation: costly clearance and persistent imbalance both change which schedules perform well. The message-content and movement controls provide further comparisons once the timing reference has been chosen.

Limitations. The timing guarantee concerns the next switch and each axis's uninterrupted interval without green in a two-axis junction. Queueing can leave an individual agent waiting beyond that interval, and the guarantee provides no ordering of arbitrary closed-loop trajectories by information quality. The workloads cover a single fixed-route junction with a reliable status interface and omit road networks, downstream spillback, report transport delay, and status-interface failures. Network stability and optimal signal timing remain outside the analysis. BG-D($4C$) ties duration to the report clock to match the original fallback in the intervention; operational deadline selection remains a separate choice, sensitive to clearance cost and demand balance.

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A Implementation and selection details

A.1 Step order, reports, and measurements

Return journeys finish at the start of a step. A ready agent joins its assigned input buffer, whose head enters at distance 48 when that cell is empty; insertion consumes no movement step. Reports describe this pre-movement state and arrive before the phase decision, with no further transport delay. Expiration and removal of observed crossings precede report insertion. Phase changes precede movement, and agents read the status interface every step. The original cycle-based rules make their first scheduled decision after one complete reporting cycle. BG-D checks the deadline every environment step, including steps without reports. A phase starting at t_0 clears during $t_0, \ldots, t_0 + c - 1$ and first opens at $t_0 + c$.

Accepted progress before time t is $P_i(t)$, so online waiting is $w_i(t) = t - b_i - P_i(t)$. Completion requires 48 accepted moves. Progress history resets at each visit and is padded with zeros until C steps are available. Agents begin each realization in return transit with remaining durations drawn uniformly from zero to 96; initial phases are balanced across the four workloads. The empirical p95 is the smallest observed value whose empirical cumulative probability reaches 0.95.

Byte 0 stores $z_i \in \{-127, \ldots, 127\}$; -128 is unused. Byte 1 is $128\mathbf{1}\{s_i = +1\} + d_i$, with $1 \le d_i \le 8$ in the reporting region. Each eligible agent attempts one report per boundary, without acknowledgment or retransmission. Offered uplink payload is $16/C$ bits per eligible agent per step, excluding identity, sequence, framing, and channel-access overhead. Fixed timers need no uplink. The common status interface carries phase, clearance, and an eight-bit gain field in two bytes per step; that field is zero when unused.

Return and route uniforms are indexed by agent and visit; erasure and action uniforms are indexed by agent and environment step. We share these arrays across paired methods and, within each population, across p and C. The shared indices fix latent draws while arrival times evolve with each controller's service history. For greedy policies, the 20 blocks contain workload variation only.

A.2 Controller selection

Test seeds are 0 through 19 and development seeds 100 through 103. Each candidate uses four workloads per development seed. Selection minimizes mean p95 subject to throughput no lower than the $4C$-step reference under the same movement policy; ties prefer fewer switches, then the lexicographically smaller parameter tuple. We retain the selected settings across the test grid and all additional workload and channel tests.

The physical-time candidates are $H \in \{6, 9, 12, 15, 18, 24, 30, 36, 48\}$, selecting 12 separately for greedy and stochastic PPO. BG-D repeats the same H search with its inherited BG evidence rule and selects 12 under greedy movement. BG-D($4C$) and the weak-cache patch are diagnostic interventions: neither receives additional tuning. The training reference selects four cycles from $\{2, 4, 6, 8\}$ at development time. BG searches minimum-hold base $m_0 \in \{1, 2\}$, stretch $\lambda_m \in \{0, 2\}$, base threshold $\theta_0 \in \{0.08, 0.12\}$, and support penalty $\theta_1 \in \{0, 0.20\}$. The minimum is $\max\{m_0 + \lceil \lambda_m(1-u) \rceil, \lceil c/C \rceil + 1\}$ and the evidence test is $ua_S > \theta_0 + \theta_1(1-u)$. Both movement policies select $(2, 2, 0.12, 0.20)$; the eight-cycle maximum and four-cycle fallback are fixed.

Packet-pressure searches $m \in \{2, 3, 4\}$, $\theta \in \{0.04, 0.08, 0.12, 0.24, 0.44, 0.64\}$, and $M \in \{4, 6, 8\}$, with $m \le M$. It selects $(3, 0.24, 6)$. Its minimum also respects $\lceil c/C \rceil + 1$. The narrow comparison searches $m \in \{1, 2\}$ and $\theta \in \{0.08, 0.12\}$ with $M = 8$, selecting $(2, 0.12, 8)$.

Each report-content control repeats the BG search and replaces the denominator offset 1 in (3) by a tuned $\kappa \in \{0.25, 0.5, 1, 2, 4\}$. The selected phase parameters remain $(2, 2, 0.12, 0.20)$. The residual and distance-weighted presence select $\kappa = 1$; unweighted presence selects 2. Presence has constant score factor one, so no scalar quantization is required. The retained dummy byte makes its erasure model identical in packet size to the residual version.

A.3 Learned movement and the optional gain

The actor and a separate value network each have two 64-unit tanh layers. The actor observes an approach-oriented 5×5 occupancy patch, distance divided by 48, a four-component approach identifier, served-axis and clearance indicators, waiting clipped at 256 and divided by 256, and a last-progress indicator. Off-lane patch cells are occupied. Buffer and return-transit agents do not act. The actor has no queue-report input.

Training uses PPO [5] with generalized advantage estimation [6]. Eight environments supply 128-step rollouts, for 1,024 joint steps per update and 400 updates per seed. Each update uses four epochs and minibatches of 4,096 on-grid transitions, retaining a smaller final batch. Environments reset after 4,096 joint steps and sample (N, C) uniformly from the nine training settings, with a $4C$-step phase. GAE ends at crossing and bootstraps at time-limit truncation. Feature and value normalization are frozen after training.

The common reward is $r_t = X_t - 0.01B_t$, where X_t counts crossings and B_t counts buffer or approach agents making no accepted progress. Adam's learning rate is 3×10^{-4}; discount and GAE parameters are 0.99 and 0.95. PPO clipping, entropy coefficient, value-loss coefficient, and gradient-norm limit are 0.20, 0.01, 0.50, and 0.50. Training uses neither BG nor a movement gain.

Full BG computes $g = uQ_S/(Q_{+1} + Q_{-1} + 1)$ after phase selection and holds $\tilde{g} = \text{round}(255g)/255$ until the next boundary. The move-logit shift is

$$\delta_i = 0.4\tilde{g}\, e^{-(d_i-1)/3}\mathbf{1}\{s_i = S,\ 1 \le d_i \le 4,\ \text{gate not clearing}\}. \tag{11}$$

The wait logit is unchanged, and the same action mask applies. The fixed-bias control replaces $0.4\tilde{g}$ with $b = 0.20$, selected from $\{0.05, 0.10, 0.20, 0.30, 0.40\}$. Deterministic PPO chooses the larger legal logit, breaking a tie toward movement.

A.4 Phase diagnostics and channel comparisons

Phase-duration histograms include phases starting in the scoring window and follow each to its terminating switch. Each realization's bin counts are normalized before averaging within and across blocks. A phase's duration includes its initial clearance. The support fractions in Table 3 use report boundaries after cache update and before phase selection, including boundaries at which the phase is held.

Switch reasons are mutually exclusive. For original BG, an empty-cache switch is labeled fallback; otherwise the eight-cycle maximum takes precedence over the evidence test. For BG-D, the physical deadline takes precedence over evidence. A suppressed fallback is recorded when the original phase reaches age $4C$ and is held. Its cache is necessarily nonempty. The weak subset additionally satisfies $n_{\text{eff}} \le 32/15$. The patch changes the empty-cache test to $u \le 4/15$ at $h \ge 4$ and leaves the other branches unchanged.

Unused green is the fraction of non-clearance steps with at least one agent at distance one on the unserved axis and none at distance one on the served axis. We compute it offline from the underlying traffic state. Per-axis waiting follows the same visit-cohort and realization-level quantile definitions as aggregate waiting.

For the asymmetric channel, each realization is evaluated twice with the erasure probabilities 0.60 and 0.80 exchanged between axes. The two label assignments are averaged within the realization before block resampling. Routing and return inputs are unchanged by the exchange. Report-independent controllers have identical trajectories under the two assignments, so their lower-loss and higher-loss axis means are exactly equal after averaging. The sensitivity tests in Table 5 change one factor at a time from the stress condition and use the same 20 blocks and four realizations per block.

B Exact single-reporter calculation

Write $D_h = 1$ when the report at cycle h is delivered, with independent probability q. Before the first switch the sole agent remains at the opposite gate. The cache at cycle h is nonempty exactly when $D_h = 1$ or $D_{h-1} = 1$. Its support never exceeds $1/8$, so the imbalance branch is inactive for every residual value.

Let $T = \tau/C$. Then T is the first $h \in \{4, 5, 6, 7\}$ with $D_{h-1} = D_h = 0$, or 8 if there is no such h. For $k = 4, 5, 6, 7$, the event $T > k$ requires that the string $D_3, \ldots, D_k$ contain no adjacent losses. Hence $\Pr(T > k) = f_{k-2}(q)$ and the tail-sum identity gives $\mathbb{E}[T] = 4 + f_2 + f_3 + f_4 + f_5$.

For $n \geq 2$, a loss-free-adjacency string either begins with a delivery, followed by a valid string of length $n - 1$, or begins with a loss and a delivery, followed by a valid string of length $n - 2$. These disjoint cases give the recurrence in (7). Turning any loss in a fixed string into a delivery can only remove an empty-cache opportunity, so T is nondecreasing under that change. Coupling delivery indicators with common uniforms proves that its expectation is nondecreasing in q. The curve in Figure 1 evaluates the exact recurrence; enumerating all 2^8 eight-cycle delivery strings gives the same values.

C Message-content and learned-movement controls

We examine the report residual and optional movement gain within the original BG scheduling family. These secondary comparisons hold the scheduling family fixed while varying the message contents or movement policy.

The residual in (1) saturates at one when $w_i \geq 64(1 + \bar{v}_i)$. A stationary agent reaches saturation after 64 waiting steps; even at maximal recent progress, saturation occurs by 128. A saturated report contributes a freshness- and distance-weighted count. Averaged over the nine loss-interval settings at each population, 57.6%, 66.9%, and 74.8% of attempted reports are saturated at populations 100, 120, and 140. We measure these fractions before quantization, across all attempts regardless of delivery.

Distance-weighted presence sets every residual to one; unweighted presence also removes the distance factor. We give each control its own development search and retain a dummy byte to match payload size. In the stress workload, distance-weighted presence changes p95 by +0.5 steps (Table 6); the grid-wide difference is +0.3, with interval $[-0.1, 0.7]$.

Table 6. Secondary controls at $N = 140$, $p = 0.70$, $C = 6$, with the original BG scheduling family. The first panel uses greedy movement and compares report contents; the second compares movement rules. Each panel has its own reference. Throughput is per 1,000 steps. Changing movement or report contents also changes later closed-loop reports and phase choices.

Control	p95 wait	Throughput	Difference [95% interval]
Report contents, with separate development tuning			
Waiting-progress residual	173.4	678.9	Reference
Distance-weighted presence	173.9	678.5	+0.5 [−0.2, 1.2]
Unweighted presence	175.2	675.4	+1.8 [0.5, 3.1]
Movement, with the residual-based scheduler			
Stochastic PPO, no gain	183.1	658.8	Reference
Stochastic PPO, report gain	180.0	664.5	−3.1 [−4.2, −2.1]
Stochastic PPO, fixed bias	179.3	665.2	−3.8 [−5.2, −2.5]
Deterministic PPO, no gain	174.1	677.6	−9.0 [−13.4, −4.4]
Greedy movement	173.4	678.9	−9.7 [−14.4, −4.9]

For stochastic PPO, a fixed near-gate bias performs similarly to the report-conditioned gain: the direct report-gain-minus-fixed-bias difference is +0.7 waiting steps, with interval $[-0.3, 1.7]$. Removing action sampling has a larger effect. Deterministic PPO reaches 174.1 steps, close to greedy movement's 173.4, suggesting that avoidable hesitation matters in this fixed-route task. These controls concern movement through an assigned phase; methods such as PressLight learn phase selection itself [4].

With the stochastic actor, the reference timer again changes the comparison (Figure 3). Full BG improves stress-case p95 from 197.9 to 180.0 relative to the 24-step training schedule. Holding the actor fixed and using a 12-step timer brings p95 to 168.4, giving a BG-minus-timer difference of +11.6 steps with interval $[9.4, 13.7]$. Throughputs are 632.5 for the 24-step timer, 664.3 for the 12-step timer, and 664.5 for full BG; the

full-BG-minus-12-step difference is +0.2, with interval $[-1.6, 2.0]$. Increasing the reporting interval also lengthens the $4C$ reference, coupling changes in communication frequency with changes in the comparison schedule.

Under the axis-dependent erasure test, full BG with stochastic PPO gives lower-loss and higher-loss p95 values of 179.3 and 216.7 steps. The lower-loss axis accounts for 50.2% of new visits and 58.2% of non-clearance service time. The greedy comparisons in Table 4 remove the movement gain from that channel-bias test.

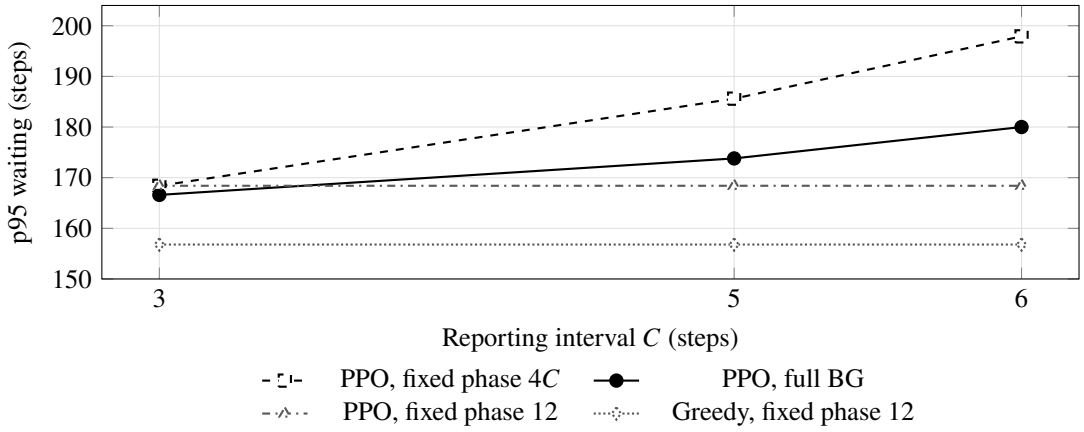


Figure 3. The reference timer changes the apparent value of communication. At $N = 140$ and $p = 0.70$, full BG's advantage over the $4C$-step reference grows as that reference becomes slower. The report-independent timers stay at the same operating points. Points are block means; Appendix C gives the corresponding stress-case controls.